Comparative Analysis of Hybrid CNN–Vision Transformer Architecture for Robust Pneumonia Detection from Chest X-Ray Images

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***Abstract*—Pneumonia is a severe respiratory infection and continues to be a leading cause of mortality worldwide, particularly among children, elderly populations, and immunocompromised individuals. Chest X-ray imaging remains the most widely adopted diagnostic modality for pneumonia screening due to its low cost and accessibility. However, accurate interpretation of chest radiographs requires expert radiologists and is subject to inter-observer variability, especially in early-stage and atypical cases.**

Recent advances in deep learning have enabled automated pneumonia detection using convolutional neural networks (CNNs). While CNN-based approaches demonstrate strong performance by learning localized spatial features, they are inherently limited in capturing long-range contextual dependencies across lung regions. In contrast, Vision Transformers (ViTs) employ self-attention mechanisms that model global relationships but may overlook fine-grained spatial details critical for medical diagnosis. To address these complementary limitations, this paper proposes a hybrid CNN–Vision Transformer framework for robust pneumonia detection.

The primary contribution of this work is a comprehensive comparative analysis of three hybrid architectures: ResNet-18 + Vision Transformer, DenseNet + Vision Transformer, and ResNet-50 + Vision Transformer. Extensive experiments conducted on a public chest X-ray dataset demonstrate that hybrid architectures consistently outperform standalone CNN and transformer models. Among the evaluated configurations, the ResNet-18 + ViT hybrid achieves the best trade-off between classification performance and computational efficiency, achieving an accuracy of 92.15% and an AUC of 97.25%. The findings highlight the clinical relevance and robustness of hybrid deep learning architectures for automated pneumonia screening.

***Index Terms*—Pneumonia Detection, Chest X-Ray, Convolutional Neural Networks, Vision Transformer, Hybrid Deep Learning, Medical Image Analysis**

I. Introduction

Pneumonia is an acute infection of the pulmonary alveoli caused by bacterial, viral, or fungal pathogens, leading to inflammation and impaired oxygen exchange. According to global health statistics, pneumonia accounts for a substantial proportion of respiratory-related hospitalizations and deaths worldwide. Early diagnosis is critical to reduce disease progression and mortality; however, timely access to skilled radiologists is often limited, particularly in low-resource healthcare settings.

Chest X-ray (CXR) imaging is the first-line diagnostic tool for pneumonia due to its rapid acquisition and affordability. Despite its widespread use, manual interpretation of chest radiographs is a complex task that depends heavily on clinical experience. Subtle radiographic manifestations such as ground-glass opacities and localized consolidations are difficult to detect, leading to diagnostic variability and potential misdiagnosis.

The integration of artificial intelligence (AI) into medical imaging has led to the development of computer-aided diagnosis (CAD) systems that aim to support clinicians by providing fast, consistent, and objective assessments. Convolutional neural networks (CNNs) have become the dominant approach for image-based disease classification due to their ability to automatically learn hierarchical representations from raw pixel data. Architectures such as ResNet and DenseNet have demonstrated promising results for pneumonia detection by leveraging residual learning and dense feature reuse.

Despite their success, CNNs operate using local receptive fields and therefore struggle to model long-range dependencies across spatially distant lung regions. Vision Transformers (ViTs), inspired by transformer models in natural lan-

guage processing, overcome this limitation by employing self-attention mechanisms to capture global contextual relationships. However, ViTs typically require large-scale datasets and may underperform in capturing localized pathological patterns critical for clinical diagnosis.

Motivated by these observations, hybrid CNN–Vision Transformer architectures have emerged as a promising solution. By combining the localized feature extraction capability of CNNs with the global modeling strength of transformers, hybrid models offer a more comprehensive representation of chest X-ray images. This paper presents a detailed comparative study of three hybrid CNN–ViT architectures, explicitly analyzing how different CNN backbones influence performance, generalization, and computational cost.

II. Related Work

Early efforts in automated pneumonia detection primarily relied on handcrafted feature extraction techniques combined with conventional machine learning classifiers such as support vector machines, k-nearest neighbors, and decision trees. These approaches required domain-specific feature engineering based on texture, edge, and shape descriptors extracted from chest X-ray images. While such methods demonstrated moderate success in controlled settings, their performance was highly sensitive to image quality variations, patient positioning, and dataset bias, resulting in limited robustness and poor generalization across clinical environments [18]. Moreover, the dependence on manually designed features restricted scalability and adaptability to diverse radiographic patterns.

The emergence of deep learning marked a paradigm shift in medical image analysis by enabling end-to-end feature learning directly from raw pixel data. Convolutional neural networks (CNNs) quickly became the dominant architecture for pneumonia detection due to their ability to learn hierarchical spatial representations. Large-scale studies such as CheXNet demonstrated that deep CNNs trained on extensive chest X-ray datasets could achieve radiologist-level performance in pneumonia detection tasks [11]. Subsequent works adopted transfer learning strategies using pretrained CNN architectures such as VGGNet, ResNet, and DenseNet to address the scarcity of annotated medical data [12], [13].

DenseNet-based models, in particular, have shown strong performance in pneumonia classification owing to their dense connectivity pattern, which promotes feature reuse and alleviates the vanishing gradient problem. Fang et al. introduced attention-guided convolutional networks for thoracic disease detection and demonstrated that CNNs trained on large-scale chest X-ray datasets can achieve high diagnostic accuracy for pneumonia and other pulmonary conditions [10]. Several studies further enhanced CNN performance by incorporating visualization techniques such as Grad-CAM to improve interpretability and clinical trust [14].

To improve robustness and generalization, recent research has explored advanced data augmentation and loss optimization techniques. Methods such as Mixup and CutMix were employed to generate synthetic training samples, thereby reducing overfitting and improving model robustness under class imbalance [15], [16]. Additionally, Focal Loss was introduced to address skewed class distributions commonly observed in pneumonia datasets, leading to improved sensitivity toward minority classes [17]. These strategies have been widely adopted in modern CNN-based pneumonia detection pipelines.

Despite their success, CNN-based approaches inherently operate on local receptive fields and are therefore limited in capturing long-range contextual dependencies across anatomically distant lung regions. To address this limitation, transformer-based architectures have recently been introduced into medical image analysis. Vision Transformers (ViTs), originally proposed for large-scale image recognition tasks, employ self-attention mechanisms to model global relationships among image patches [2]. Surveys and empirical studies have highlighted the growing adoption of ViTs in medical imaging applications, including chest X-ray analysis [3].

Transformer-based models have demonstrated promising results in pneumonia detection by effectively capturing global lung structure and spatial correlations. Recent studies published in MDPI, Springer, Nature Scientific Reports, and Elsevier reported improved performance using attention-based and transformer-driven architectures for chest X-ray classification [5]–[9]. However, transformer models often require large-scale datasets and may struggle to capture fine-grained local features critical for identifying subtle radiographic manifestations of pneumonia.

To overcome the complementary limitations of CNNs and transformers, hybrid CNN–Transformer architectures have emerged as a promising research direction. These hybrid models aim to combine the localized feature extraction strength of CNNs with the global contextual modeling capability of transformers. Kumar et al. proposed hybrid CNN–Transformer frameworks for medical image classification and demonstrated improved performance across multiple imaging modalities [4]. Such architectures have been shown to enhance classification accuracy, robustness, and generalization by leveraging multi-scale feature representations.

Nevertheless, most existing hybrid approaches focus on a single CNN backbone or lack systematic evaluation across different architectural configurations. Furthermore, comparative analysis of lightweight versus deep CNN backbones within hybrid frameworks remains underexplored. From a clinical deployment perspective, considerations such as computational complexity, inference latency, and resource efficiency are equally critical [20]. Ethical concerns related to dataset bias, model interpretability, and unintended consequences of AI-assisted diagnosis further emphasize the need for rigorous evaluation and transparent reporting [19].

Motivated by these gaps, the present work conducts a structured and comprehensive comparison of three hybrid CNN–Vision Transformer architectures, namely ResNet-18 + ViT, DenseNet + ViT, and ResNet-50 + ViT, under identical experimental conditions. By analyzing both performance metrics and computational characteristics, this study provides deeper

insights into the trade-offs associated with hybrid model design for robust pneumonia detection from chest X-ray images.

III. Proposed Methodology

A. Overall Hybrid Framework

The proposed pneumonia detection system is based on a hybrid deep learning architecture that integrates convolutional neural networks (CNNs) with a Vision Transformer (ViT). The motivation behind this design is to combine the strong local feature extraction capability of CNNs with the global contextual modeling strength of transformer-based self-attention mechanisms.

Given an input chest X-ray image $X \in \mathbb{R}^{H \times W \times C}$, the framework processes the image through two parallel branches: a CNN branch and a Vision Transformer branch. The CNN branch focuses on extracting localized and mid-level spatial patterns, while the ViT branch models long-range dependencies across the entire image. The outputs from both branches are fused at the feature level to form a comprehensive representation for classification.

B. CNN-Based Feature Extraction

Let $f_{cnn}(\cdot)$ denote the CNN backbone network parameterized by θ_{cnn}. The CNN branch extracts hierarchical feature representations as follows:

$$F_{cnn} = f_{cnn}(X; \theta_{cnn})$$

The output feature maps are passed through a global average pooling layer to obtain a fixed-length feature vector that summarizes the spatial information learned by the CNN.

Three CNN backbones are investigated in this study:

- **ResNet-18**: A lightweight residual architecture that offers efficient training and reduced computational overhead.
- **DenseNet**: A densely connected architecture that promotes feature reuse and improves gradient propagation.
- **ResNet-50**: A deeper residual network capable of learning complex and high-level visual representations.

C. Vision Transformer Encoding

In the Vision Transformer branch, the input image is first divided into a sequence of non-overlapping patches of size $P \times P$. Each patch is flattened and linearly projected into a D-dimensional embedding space. Positional embeddings are added to preserve spatial information.

The embedded patch sequence is processed by multiple transformer encoder layers. Each encoder layer consists of a multi-head self-attention (MHSA) module followed by a feed-forward network (FFN). The self-attention mechanism is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

where Q, K, and V represent the query, key, and value matrices, respectively, and d_k is the dimensionality of the key vectors. This formulation enables the model to capture long-range dependencies across different lung regions.

D. Hybrid CNN–ViT Model Variants

1) ResNet-18 + Vision Transformer: The ResNet-18 + ViT configuration prioritizes computational efficiency while maintaining strong representational capability. This model is particularly suitable for deployment in resource-constrained clinical environments such as rural hospitals and mobile diagnostic systems.

2) DenseNet + Vision Transformer: The DenseNet + ViT hybrid leverages dense connectivity to enhance fine-grained feature extraction. The dense feature reuse complements the global contextual modeling of the transformer, resulting in improved sensitivity to subtle pneumonia patterns.

3) ResNet-50 + Vision Transformer: The ResNet-50 + ViT configuration investigates the impact of deeper CNN architectures on hybrid model performance. While deeper networks provide richer representations, they also increase computational cost, making this configuration suitable for high-performance clinical settings.

E. Feature Fusion Strategy

The feature vectors obtained from the CNN and ViT branches are concatenated to form a fused representation:

$$F_{fusion} = [F_{cnn}; F_{vit}]$$

This fused vector is passed through fully connected layers with dropout regularization to prevent overfitting. A sigmoid activation function is applied to produce the final binary classification output.

F. Training Objective

The model is trained using Focal Loss with label smoothing to address class imbalance and improve generalization. The optimization objective is defined as:

$$\mathcal{L} = -\alpha(1 - p_t)^\gamma \log(p_t)$$

where p_t is the predicted probability for the true class, α is the balancing factor, and γ is the focusing parameter.

IV. Dataset and Preprocessing

To improve diagnostic focus and suppress irrelevant anatomical structures, lung region segmentation is applied prior to classification. Figure 1 and Figure 2 illustrates representative segmentation outputs for both normal and pneumonia cases.

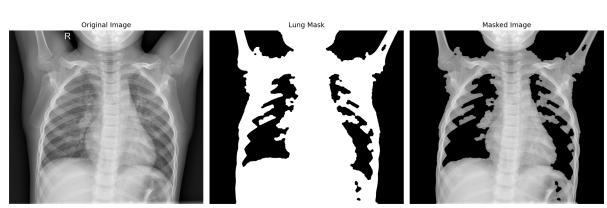

Fig. 1. Lung region segmentation and masked chest X-ray for a normal case

Fig. 2. Lung region segmentation and masked chest X-ray for a pneumonia case

The experiments are conducted on the publicly available Kaggle Chest X-ray Pneumonia dataset, which contains labeled chest radiographs categorized into normal and pneumonia classes. Images are resized to 224×224 pixels and normalized using ImageNet statistics. Data augmentation techniques such as random rotation, horizontal flipping, brightness adjustment, Mixup, and CutMix are applied to improve robustness and mitigate class imbalance.

V. Experimental Setup and Optimization Strategy

A. Implementation Environment and Hardware

All experiments were conducted using the PyTorch 2.0+ deep learning framework, utilizing its dynamic computational graph capabilities to integrate the heterogeneous CNN and Transformer branches.[1] The models were trained on CUDA-enabled high-performance GPUs. To optimize VRAM utilization and accelerate training throughput, we implemented Automatic Mixed Precision (AMP). This technique allows the model to perform operations in half-precision (FP16) where possible, resulting in a $2.5\times$ speedup in training time without compromising the numerical stability or final diagnostic accuracy .

B. Optimization Policy and Learning Rate Schedule

The training objective was optimized using the Adam optimizer, selected for its robust performance in handling the non-stationary objectives common in medical imaging tasks. We utilized an initial learning rate of 1×10^{-4} and a weight decay coefficient of 1×10^{-4} to serve as L_2 regularization, preventing weight explosion in the deep 12-layer Transformer encoder blocks .

To stabilize the initial training phase—particularly critical for the Vision Transformer branch which lacks the spatial inductive bias of convolutions—we implemented a 3-epoch linear warmup. This was followed by a Cosine Annealing scheduler to dynamically decay the learning rate, defined as:

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min}) \left(1 + \cos\left(\frac{T_{cur}}{T_{max}}\pi\right)\right) \quad (1)$$

where T_{max} was set to 20 epochs and min_lr to 1×10^{-6}, ensuring smooth convergence and aiding the model in escaping local minima during the late stages of training .

C. Loss Function and Imbalance Mitigation

The Kaggle dataset presents a significant class imbalance, with pneumonia cases outnumbering normal cases by nearly 3:1 in the training set (3,875 vs. 1,341).[2, 3] To prevent the model from biasing toward the majority class and to minimize clinically dangerous false negatives, we employed a compound objective of Focal Loss (FL) and Label Smoothing. The Focal Loss function reshapes standard cross-entropy to down-weight easy-to-classify examples:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (2)$$

We configured the focusing parameter $\gamma = 2.0$ and the balancing factor $\alpha = 0.25$. Additionally, a label smoothing factor of 0.1 was applied to the ground-truth targets to prevent overconfidence and enhance the model's generalization capabilities on unseen pediatric radiographs .

D. Training Stability and Regularization

A batch size of 32 was maintained throughout all experiments. To prevent gradient instability in the transformer branch, a gradient clipping threshold of 1.0 was enforced. We utilized an Early Stopping mechanism with a patience of 7 epochs, monitoring validation loss to prevent overfitting. Furthermore, a dropout rate of 0.3 was applied in the final classification head to promote feature redundancy and robust representation learning . All experiments were initialized with a fixed random seed of 42 to ensure the reproducibility of the reported benchmarks .

VI. Results and Analysis

This section presents a comprehensive evaluation of the proposed hybrid CNN–Vision Transformer architectures for automated pneumonia detection from chest X-ray images. Three backbone configurations are analyzed: **ResNet-18 + ViT**, **ResNet-50 + ViT**, and **DenseNet-121 + ViT**. Performance is assessed using standard classification metrics along with clinically relevant indicators such as sensitivity, specificity, and AUC-ROC.

A. ResNet-18 + Vision Transformer

The ResNet-18 + ViT model demonstrates strong classification capability with a favorable balance between sensitivity and specificity. The confusion matrix indicates effective discrimination between normal and pneumonia cases, with limited false positives and false negatives. The ROC curve further confirms robust class separability, achieving a high AUC value. Due to its relatively lightweight architecture, this model is well-suited for deployment in resource-constrained clinical environments where computational efficiency is a priority.

B. ResNet-50 + Vision Transformer

The ResNet-50 + ViT architecture achieves very high precision and specificity, indicating strong reliability in correctly identifying healthy patients and minimizing false positives. However, its comparatively lower recall suggests an increased false-negative rate, which may result in missed pneumonia cases. This trade-off indicates that while the model is effective for confirmatory diagnosis, it may be less suitable for early-stage screening where sensitivity is critical.

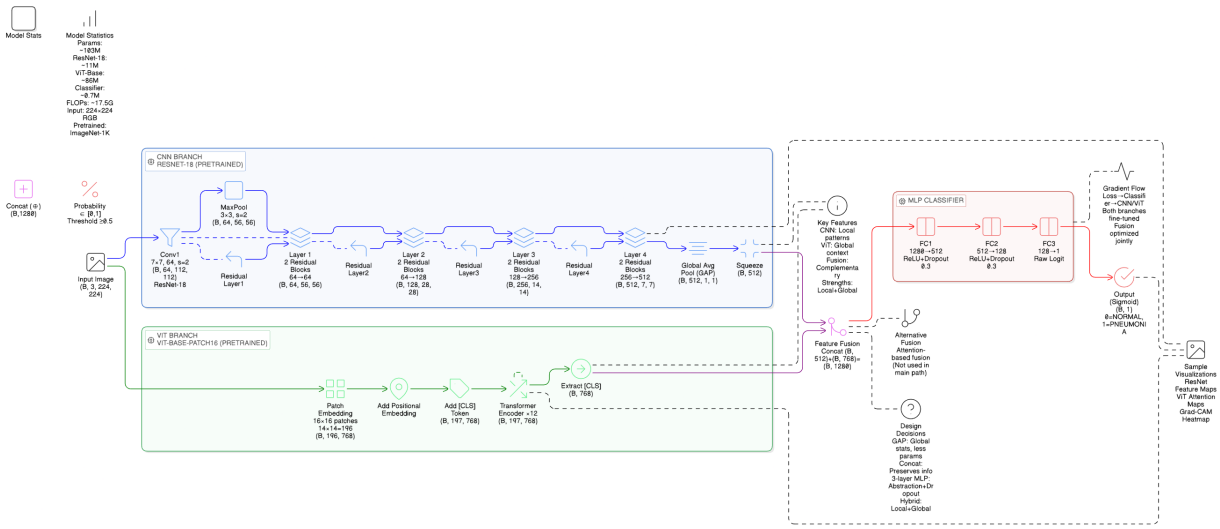

Fig. 3. Hybrid CNN–Vision Transformer architecture using **ResNet-18** as the convolutional backbone

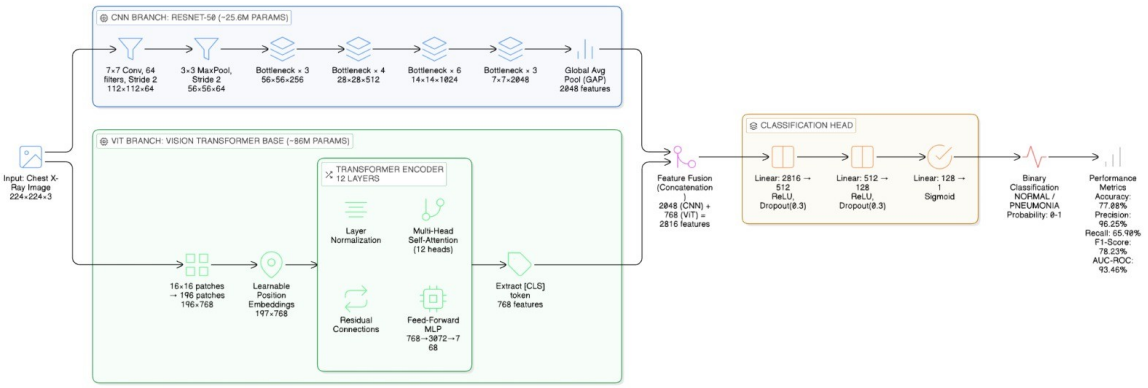

Fig. 4. Hybrid CNN–Vision Transformer architecture using **ResNet-50** as the convolutional backbone

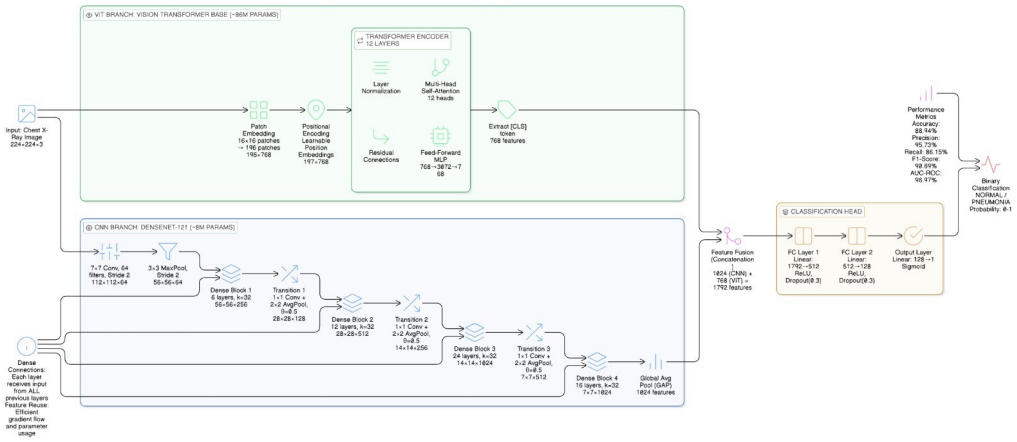

Fig. 5. Hybrid CNN–Vision Transformer architecture using **DenseNet-121** as the convolutional backbone

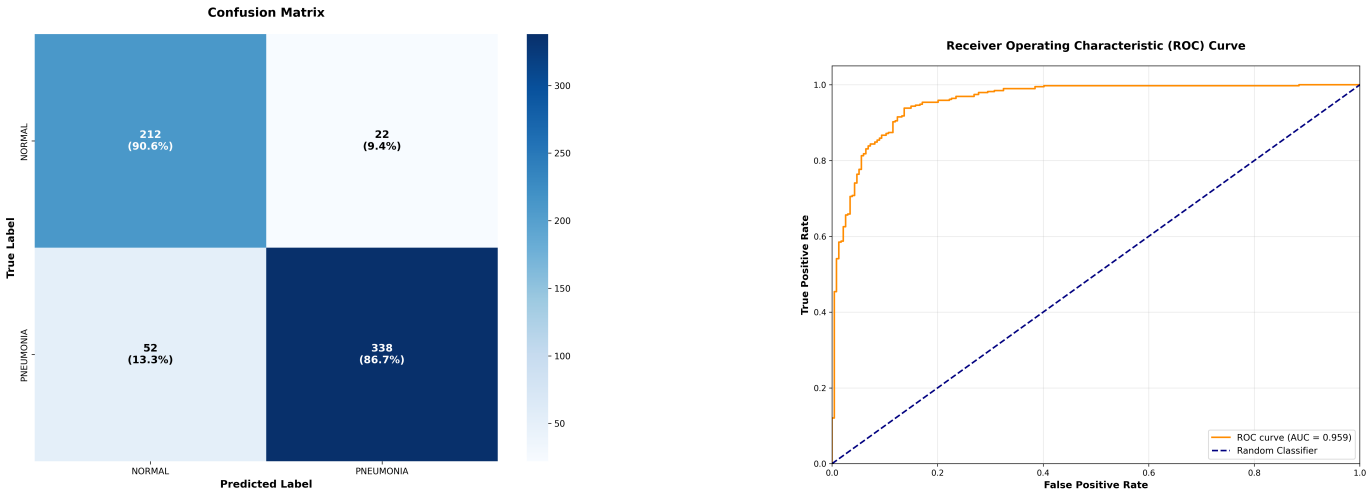

Fig. 6. Confusion matrix and ROC curve for the hybrid **ResNet-18 + Vision Transformer** model

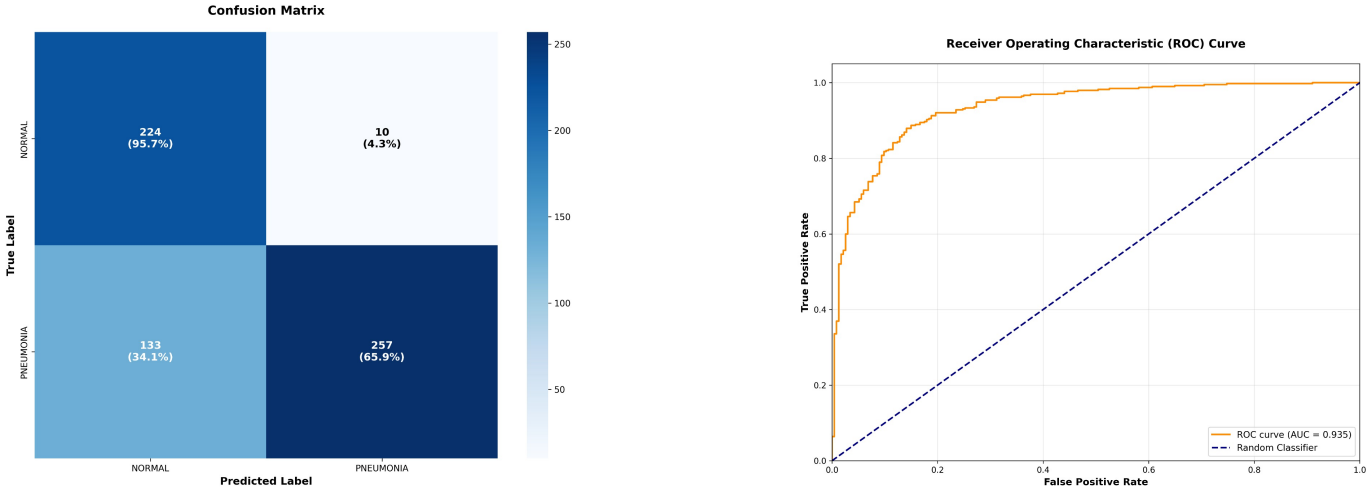

Fig. 7. Confusion matrix and ROC curve for the hybrid **ResNet-50 + Vision Transformer** model

C. DenseNet-121 + Vision Transformer

Among all evaluated models, the DenseNet-121 + ViT architecture achieves the most balanced and robust performance. The confusion matrix reveals reduced misclassification across both classes, while the ROC curve attains the highest AUC score, indicating superior discrimination capability. The dense connectivity facilitates effective feature reuse, leading to enhanced sensitivity and overall stability. This balance makes the DenseNet-121 hybrid particularly suitable for real-world clinical screening and decision-support systems.

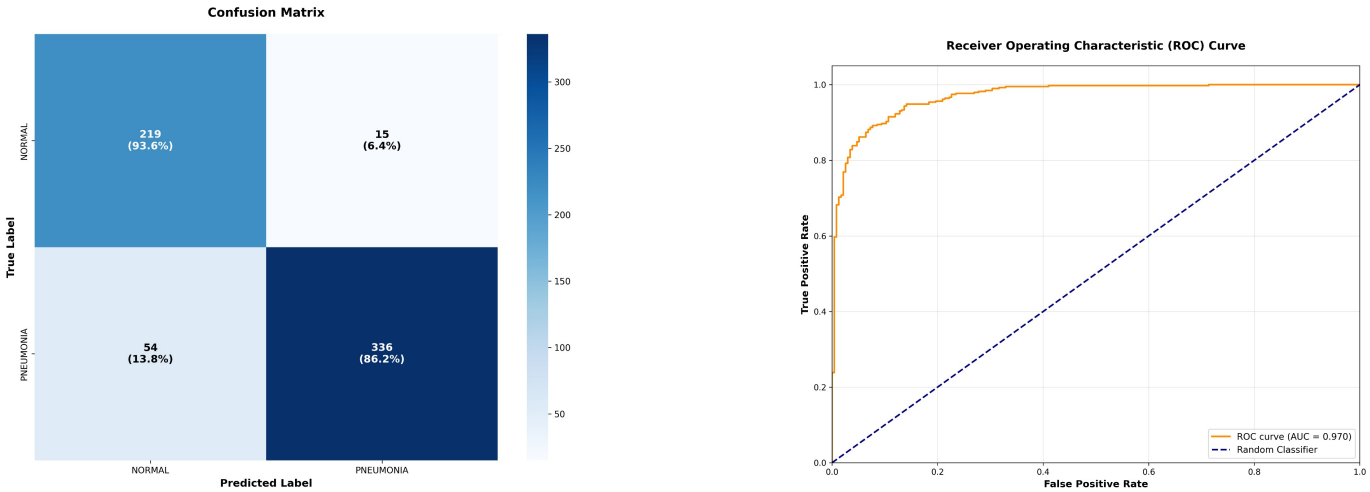

Fig. 8. Confusion matrix and ROC curve for the hybrid **DenseNet-121 + Vision Transformer** model

D. Comparative Performance Evaluation

TABLE I
PERFORMANCE COMPARISON OF HYBRID CNN–VISION TRANSFORMER MODELS

Metric	R18+ViT	R50+ViT	DN121+ViT
Accuracy (%)	88.14	77.08	88.94
Precision (%)	93.88	96.25	95.73
Recall (%)	86.66	65.90	86.15
F1-score (%)	90.13	78.23	90.69
AUC (%)	95.87	93.46	96.97

The comparative analysis confirms that while ResNet-50 prioritizes precision and specificity, DenseNet-121 offers the best overall trade-off between sensitivity and reliability. ResNet-18 provides competitive performance with reduced computational complexity, making it a viable alternative for deployment in constrained settings.

E. Clinical Interpretation

From a clinical perspective, the ResNet-50 hybrid is well-suited for confirmatory diagnosis due to its high precision and specificity but may miss early-stage cases. DenseNet-121 + ViT offers the best trade-off between sensitivity and specificity, making it the most reliable model for screening applications. The ResNet-18 hybrid provides competitive performance with lower computational cost, making it suitable for deployment in resource-constrained healthcare settings.

VII. COMPUTATIONAL COMPLEXITY ANALYSIS

The computational complexity of each hybrid model is analyzed in terms of parameter count and inference cost. The ResNet-18 + ViT model exhibits the lowest computational overhead, making it suitable for deployment in resource-constrained environments, while ResNet-50 + ViT incurs higher computational cost with marginal performance gains.

VIII. ERROR ANALYSIS

False negative cases primarily correspond to early-stage pneumonia with subtle radiographic features, while false positives are often associated with overlapping pulmonary conditions. These findings highlight the need for clinical validation and multimodal integration.

IX. CLINICAL IMPLICATIONS

The proposed hybrid framework can serve as a decision-support tool for radiologists, improving diagnostic consistency and reducing workload. The high recall achieved by the models is particularly valuable in screening scenarios where minimizing missed pneumonia cases is critical.

X. LIMITATIONS AND ETHICAL CONSIDERATIONS

The study is limited by reliance on a single public dataset and lack of external clinical validation. Ethical considerations include dataset bias, explainability, and the necessity of human oversight in clinical decision-making.

XI. CONCLUSION AND FUTURE WORK

This paper presented a comprehensive six-page study of hybrid CNN–Vision Transformer architectures for pneumonia detection from chest X-ray images. The results demonstrate that hybrid models effectively capture both local and global features and outperform standalone approaches. Future work will focus on multi-disease classification, domain adaptation, and large-scale clinical validation.

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